Progress Log : Weeks 3–4

1. **What I've done:**

Data Collection:

- Narrowed the focus of the study area from the entire area to a 5km radius around the University of Strathclyde, as agreed for improved focus and feasibility within the timeframe available for the study.
- Identified 7 SEPA potentially vulnerable areas within the study area including Glasgow City Centre, River Kelvin and Rutherglen.
- Downloaded and verified OpenStreetMap data 55,638 buildings, 38,218 roads and 195 water bodies.
- Obtained NASA SRTM elevation data 259,200 points at 30m resolution (better resolution than the original 400-point grid).
- Created labelled maps of the study area to show flood zones, buildings, roads, water bodies, and elevation, with maps of original vs NASA SRTM elevation data to compare.

Ethics Application:

- Compiled consent form and Prepared Participant Information Sheet following the CIS Departmental Ethics Committee template which is to be approved by the supervisor.

EDA and Dissertation:

- Performed exploratory data analysis (EDA) on all of the data collected, to be finished by the end of week.
- Started to write introduction and data sections of dissertation report.

2. **Issues or barriers:**

- The NASA GPM rainfall data revealed no spatial variation throughout the 5km study area with the same value returned by all the grid points (3.78mm/day annual average).

3. **Planning for next two weeks:**

- To Complete EDA and started feature engineering, such as calculating distance to nearest river from OSM water body data to extracting land use features.
- Perform all CRS alignment and spatial joining to generate the feature matrix with all the features labelled and ready for machine learning.